Retrieval Capabilities of Legal AI Tools

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1 Abstract

Legal technology vendors offer products with different capabilities. One capability often offered is the ability to upload documents as a custom database and query it with an AI chat functionality. This study presents an experiment, data, and analysis of the performance of these legal AI tools from Westlaw, Lexis, and Harvey with different kinds and sizes of uploaded text. I create a specified set of files and seed "clues" that correspond to questions I ask the legal AI tool covering a range of retrieval and logic tasks. The legal AI's response is evaluated on the basis of its ability to retrieve the seeded information. This analysis and data describes the relationship between legal AI performance and both file set sizes and corpora types, specifically retrieval capability and accuracy. I conclude that there are correlations between file set size and performance, and for all tested products, a correlation between corpora and performance. This research aims to provide data and analysis that users of legal AI tools can use to better understand these tools and make informed decisions about use by critically assessing the legal AI tools along the variables used in this research.

2 Background and Literature review

Electronic document management platforms have long been part of law practice and are an area that generative AI has and continuyes to disrupt. A critical element of this disruption is the ability to use legal AI tools not just to analyze documents, but to identify relevant documents out of many. This creates a new market for legal AI tools such as Harvey and Legora, which are not part of large pre-existing legal databases of authorities. Traditional legal tech platforms have also expanded to offer legal AI tools with this functionality as well.

Despite this, there is very little independent evaluation or comparison of available legal AI tools for this specific purpose. Some of this is due to lack of access to multiple tools with similar functionality. Contracts with platforms also may prohibit public disclosure of analysis or technical findings. Another factor may be the labor required to gather data at a scale that researchers can statistically analyze. Consequently, while a lot of great research has been and is continually done on the research or generative abilities of legal AI tools, there is currently very little analysis or scrutiny of the ability of these tools to interact with user-uploaded files.

The one study that I could identify addressing a similar question is the Vals Legal AI Report from February 27, 2025 by Vals AI.[1] This study covers many different legal AI tools and types of tasks that legal AI tools may be asked to do. Of the many tasks, Data Extraction and Document Q&A tests are the most similar to the focus of my study. However, this study addresses a fundamentally different research question, specifying no more than a single-digit number of files as a source for a legal AI's response. This study also confined user-uploaded files to sets of no more than 29 in any trial and only uploading documents pertaining to that specific trial. In this study's "Notes on study limitation" section, the researchers specifically acknowledge that they were limited by the nature of the type of documents and question/answer pairs in their experimental design. While the Vals study is useful in addressing the use case of users uploading small and specific sets of files, it does not examine a use case with more files of varying relevance. Given the fact that some of these legal AI tools allow a much larger capacity than what the Vals AI study tests, it leaves many potential use cases unassessed.

It is into this research gap that this research steps.

3 Methodology

All code is available at this Git Repository: https://github.com/JayTongue/Sherlock_HoLLMes

3.1 Overview:

This paper evaluates the capability of different legal AI products by grading their ability to answer questions based on a standardized set of problems seeded throughout files containing different kinds of text (corpora).

[1] Vals AI, *Vals Legal AI Report* (Feb. 2025), https://www.vals.ai/industry-reports/vlair-2-27-25 [https://perma.cc/58WN-ZS33]

I chose products which offered the ability to upload and query a custom set of files with a legal AI tool. The evaluated tools for this study were Westlaw CoCounsel, Lexis Protégé, and Harvey.

I tested five different kinds of corpora: two based on real-world data sets and three synthesized corpora. The real-world data sets were download from open-source online repositories. Synthetized data sets were algorithmically generated with a variety of computational means.

The Clues that formed the questions followed standardized templates, swapping the names of individuals, topics, and other fields so that they do not overlap and entangle, and questions always have some element of spontaneous iteration and variability. After creating files and clues, I injected the clues into random places in randomly selected files in a given set.

I then uploaded the injected file set to a legal AI tool and asked it the corresponding questions. I then evaluated the legal AI's answers and recorded the results.

3.1.1 Products Tested

The products evaluated in this study were confined to those available to the author in late 2025 and early 2026. These tools were Westlaw CoCounsel 2.0, Lexis Protégé, and Harvey.

Westlaw CoCounsel 2.0 is a legal AI tool released by Westlaw, a legal publisher and research platform owned by Thomson Reuters. Westlaw released their CoCounsel AI in August of 2025.[2]

Lexis is also a legal publisher and research database. They are owned by RELX, a company based in the United Kingdom[3]. Lexis released their legal AI product, Protégé, in January 2025 for general availability in the United States.[4] The evaluated version of Lexis Protégé does not have a specific version appociated with it, and is simply branded as "Lexis+ with Protégé".

Westlaw and Lexis are sometimes referred to as a duopoly because of their dominant market positions for offering databases of legal and law-adjacent sources.[5] While they do not face as much competition from other companies when it comes to legal databases, the area of legal AI is broadly contested, with many emerging players in the field. One such company is Harvey.

Harvey was founded in 2022 as a legal tech startup. In December of 2025, a funding round valued the company at 8 billion dollars.[6] Harvey offers a variety of legal AI products such as the ability to create custom databases, make user-defined workflows, and iteratively produce and generate drafts of legal documents. The version of Harvey evaluated in this study did not have a version number.

This study confines its scope to the ability to upload a custom database and query it with a legal AI. All three of these platforms offer this ability. Lexis and Harvey refer to these user uploaded files as a "vaults", whereas Westlaw simply refers to them as "databases". All seem to be analogous tools by a different name.

3.1.2 Data Sources and File Sets

A file set refers to the files used for a single upload and a single set of questions. Every file set is comprised of content from one of five corpora types with a specified number of files in the set. For instance, an Enron file set of size 25 would contain 25 files randomly selected from the Enron dataset.

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[2]Thomson Reuters, *Thomson Reuters Launches CoCounsel Legal: Transforming Legal Work with Agentic AI and Deep Research*, THOMSON REUTERS (August 5, 2025) https://www.thomsonreuters.com/en/press-releases/2025/august/thomson-reuters-launches-cocounsel-legal-transforming-legal-work-with-agentic-ai-and-deep-research [https://perma.cc/A5S5-PZ26] (last visited Feb. 17, 2026).

[3]RELX, *Legal*, RELX, https://www.relx.com/our-business/market-segments/legal [https://perma.cc/U9QF-BAQF] (last visited February 17, 2026).

[4]LexisNexis, *LexisNexis Introduces Protégé Personalized AI Assistant with Agentic AI, Making it Easier to Power Complex Legal Task Completion*, LEXISNEXIS (January 27, 2025) https://www.lexisnexis.com/community/pressroom/b/news/posts/lexisnexis-introduces-protege-personalized-ai-assistant-with-agentic-ai-making-it-easier-to-power-complex-legal-task-completion [https://perma.cc/77K3-QDPH] (last visted Feb. 17, 2026).

[5]See e.g. Tom Blakely, *Gatekeepers of Law: Inside the Westlaw and LexisNexis Duopoly*, BIG BY MATT STOLLER (Dec 31, 2025) https://www.thebignewsletter.com/p/gatekeepers-of-law-inside-the-westlaw [https://perma.cc/3PDN-PADX] (last visited Feb. 17, 2026).

[6]Michael J. de la Merced, *Harvey, a Maker of A.I. Legal Software, Raises New Funds*, THE NEW YORK TIMES (Dec. 4, 2025) https://www.nytimes.com/2025/12/04/business/dealbook/harvey-legal-ai.html [https://perma.cc/L222-8UHG] (last visited Feb. 17, 2026).

3.1.2.1 Corpus Types This study used five different types of corpora, meant to span a range of possible files in a legal setting. Files from the Contracts and Enron datasets were high in semantic content, whereas the synthetic corpora (Markov, Random, and Zeros) were low in semantic content. This paper uses "semantic content" to refer to content which conveys cognizable information. For instance, a clause of a contract conveys information about the agreement between the parties, whereas a string of random characters conveys no information. Besides the Contracts and Enron corpora which are semantically meaningful, the Markov copora mimics the text of legal opinions while the Random corpora mimics data which may be encrypted or corrupted, and the Zeros corpora mimics data which may be blank, overwritten, or redacted.

I included the random and zero data sets in this study to act as baselines to compare legal AI performance. Their complete lack of any human-readable language except for the clues isolates the element of pure dilution by allowing assessment of model performance without any confounding semantic content from the underlying corpus. It is against these that the performance of more realistic datasets such as those made from the contracts or enron corpora are contextually evaluated.

I included the Markov corpus because of the interesting properties of Markov text. Specifically, although Markov text is statistically valid by definition, it is semantically meaningless. Since one of the features of large language models is that they process and understand text, the Markov corpus poses an interesting comparison to truly meaningful corpora from the real world.

3.1.2.1.1 Commercial Contracts File sets of the "Contracts" filetype were filled with files randomly selected from the Material Contracts Corpus compiled in 2025 by Stanford Law School. This dataset is mostly comprised of contracts publicly available in the SEC EDGAR database. It contains the text of 1,038,766 contracts as well as metadata, totalling 156.2 GB. I downloaded this dataset from the Material Contracts Corpus website[7], indexed it, and randomly sampled it for the text of file sets of this corpus type.

3.1.2.1.2 Enron Discovery File sets of the "Enron" filetype were filled with files randomly selected from the Enron Email Dataset V2. Although this dataset has received a lot of analysis and scrutiny, it is quite difficult to find the dataset in a complete, un-filtered form. The fullest dataset available to the author was from the Internet Archive[8]. These files were downloaded, extracted, and indexed. Fully extracted, this dataset takes 258.9 GB.

A portion of this data is machine-generated text, such as email headings, signatures, and other computationally added information besides the message of the actual emails. Files for an Enron type file set contain the text of files randomly selected from this dataset.

3.1.2.1.3 Markov Text File sets of the "Markov" filetype contained text generated from Markov Chains trained on the United States Reports. Markov Chains are mathematical descriptions of the pattern of text in a training corpora. Simplistically, they record the statistical likelihood of any current textual state leading to a given future textual state.[9]

I downloaded the United States Reports (1754-2014) from the Caselaw Access Project[10]. I first normalized the text of the opinions with a normalization function called NFKC[11]. This normalization collapsed the text of the characters into a more manageable encoding space by changing uncommon characters into more common characters, reducing the number of total characters into a more manageable "canon". This eliminated variations in encoding that would skew the Markov chains by modeling unintentional or unimportant distinctions between characters. For instance, the "ﬁ" ligature defined as a single Unicode character (U+FB01) would be transformed to "fi" as two separate letters through NFKC. This step is important because the

[7] Peter Adelson and Julian Nyarko, *Material Contracts Corpus*, Stanford Law School https://mcc.law.stanford.edu/ [https://perma.cc/XC7Z-AJ7G] (last visited Feb. 17, 2026).

[8] Enron Corporation, *Files for edrm.enron.email.data.set.v2.xml*, Internet Archive https://archive.org/download/edrm.enron.email.data.set.v2.xml [https://perma.cc/HYM9-72YQ] (last visited Feb. 17, 2026).

[9] See e.g. Joseph Chang, *Markov Chains*, (February 2, 2007) http://www.stat.yale.edu/~pollard/Courses/251.spring2013/Handouts/Chang-MarkovChains.pdf [https://perma.cc/3S23-J3A3] (last visited Feb. 17, 2026) (A chapter of an unpublished textbook manuscript).

[10] United States Government Publishing Office, *United States Reports (1754-2014)*, Caselaw Access Project https://case.law/caselaw/?reporter=us [https://perma.cc/5PV9-6WMT] (last visited Feb. 17, 2026).

[11] Ken Whistler, *Unicode Normalization Forms*, Unicode Standard Annex #15 (2025-07-30) https://www.unicode.org/reports/tr15/ [https://perma.cc/Q936-SMC2] (last visited Feb. 17, 2026).

Caselaw Access Project used Optical Character Recognition (OCR) on scans of physical reporters.[12] While OCR is very useful for translating physical or other non-machine coded media into a text-encoded form, it often results in recognition errors due to imperfections in the physical source medium.

I then tokenized the text with a regular expression that split the text into a list of continuous characters and punctuation from the block of normalized text. I iterated through this tokenized text to construct 3rd-order, 2nd-order, and 1st-order markov chains.[13] These chains then generated text from a single randomly selected starting 3rd-order key. If a situation was not found as the chains generated text, the 3rd-order chain rolled back to 2nd-order, then the 1st-order chain if necessary.

Files of the Markov corporus used these chains to generate text until the text met the target file size.

3.1.2.1.4 Random Characters File sets of the "Random" filetype contained the text of characters from generated from the /dev/urandom and /dev/random files in a linux filesystem. These two files allow users to access the computer's random number generator, returning a random stream of bytes based on the computer's entropy pool.[14]

Files in the Random file sets used a stream of random bytes from these random generators until they contained the target number of bytes. While the initial experimental design was to generate files that spanned the whole range of possible byte combinations for maximum randomness, this often led to errors when trying to encode that text into some file types, especially DOCX. This meant that the characters that caused errors would either have to be removed individually or with a heuristic rule.

To curtail these issues, I regularized the data as hexadecimal strings. Although this is a limitation on the kind of characters the files could contain, this ensured that the characters written to Random files were in the file type's encodable space and would therefore convert into that file type without encoding errors.

I filled files in Random file sets with these regularized random bytes until the text met the target file size.

3.1.2.1.5 Zeros Files in the Zeros file sets contained literally repeated numerical zeros ("0"). Similarly to streamed random bytes in the Random corpus, a function streamed zeros into files until they contained the target file size.

3.1.2.2 Distribution Analysis In order to mimic a realistic file size distribution for the synthetic corpora types (Markov, Random, and Zeros), I analyzed the file size distribution of the two publicly available datasets (Contracts and Enron) so that the generated files could follow roughly the same distribution. Since I was regularizing sizes of individual files along a set distribution, I decided to use number of files in a set as a proxy metric for overall size of information. Since both the Contracts and Enron datasets could be fit to a lognormal distribution, I followed an average of their distributions when creating synthetic corpora. A less limited study could explore the effect of different types of distributions.

After analyzing the file sizes of both the Enron and Contracts datasets, I fit this file size data to a lognormal distribution:

$$X = e^{\mu+\sigma Z} \sim \mathrm{LogNormal}(\mu, \sigma)$$

where $Z \sim \mathcal{N}(0, 1)$.

Parameters:

- μ is the mean of $\ln X$
- σ is the standard deviation of $\ln X$
- $Z \sim \mathcal{N}(0, 1)$ is a standard normal random variable

[12] The Library Innovation Lab Team, *Transitions for the Caselaw Access Project*, LIBRARY INNOVATION LAB (Mar. 26, 2024), https://lil.law.harvard.edu/blog/2024/03/26/transitions-for-the-caselaw-access-project/ [https://perma.cc/K6PY-9FM5]

[13] E.g. given the text `"We the People of the..."`, an example of a 3rd-order markov chain would be for the key: `('We', 'the', 'People')` to correspond to the value: `'of'`. A 2nd-order markov chain would pair the key: `('We', 'the')` with the value `'People'`. A 1st-order markov chain would pair the key: `('We')` to the value `'the'`. Because higher orders consider more context when deciding the next value, they are generally prefered for generating more realistic text and avoiding inexcapable loops.

[14] Linux Foundation, *urandom(4) - Linux man page*, LINUX MANUAL https://linux.die.net/man/4/urandom [https://perma.cc/5CS5-V7KB] (last visited Feb. 17, 2026).

The Contracts dataset contains consistent filetypes and relatively consistent content, fitting the lognormal distribution decently. However, the Enron database's distribution was very irregular, with spikes corresponding to different types of files or contents. For instance, simple emails would tend to be one size, emails with attachments would be another size, machine log files would be another, etc. There were also files as small as 10 bytes, which were functionally useless for the purposes of this research. When calculating parameters and randomly selecting files from both the Contracts and Enron datasets, I implemented a 1KB floor to eliminate these miniscule files. I did not implement an upper limit of file size.

Metric	Contracts Dataset	Enron Dataset
μ (mu)	10.866	9.0723
σ (sigma)	1.4167	1.8434
Median	52 KB	1.2 KB
95% Range	2 KB, 900 KB	10 bytes, 10 MB
Notes	Well-behaved distribution	Heavy right tail with extreme outliers

Despite the lognormal distribution not fitting well to the Enron dataset, the lognormal distribution was the one that fit the model the best without breaking the distribution down into different functions or overfitting with a high-order polynomial. After fitting, I averaged the Mu and Sigma parameters from both datasets to create parameter targets for the synthetic data sets. These averaged parameters yielded file size targets sitting in between the Contracts and Enron datasets.

- $\mu = 9.96915$,
- $\sigma = 1.63005$

I used these parameters to generate target file sizes to be filled with Markov text, random characters, or zeros.

3.1.2.3 File Types Each commercial vendor permitted different types of files. Harvey was the most permissive, allowing PDF, DOCX, RTF, TXT, EML, MSG, XLS, XLSX, CSV, PPT, PPTX, HTML, as well as code files written in various programming languages.[15] Westlaw allows PNG, JPEG, Word, Excel, CVS, PDF, and zips of each of the other accepted file types.[16] Lexis was the most restrictive, allowing only PDF, DOC, DOCX, TXT, and ZIP files.[17]

Note that Lexis' restrictiveness, to some extent, is arbitrary. Many of the other file formats such as CSV, HTML, and EML can be opened as plain text files and do not exist in a specialized encoding that requires an additional parser. This restrictiveness is an interesting business and/or engineering decision, especially since documentation for other Lexis tools seems to show that other tools had permit more file types.[18]

However, due to this restriction, I chose to randomize files as either PDF, TXT, or DOCX in approximately equal proportion since the permitted ZIPs would just be decompressed into those file types before actual processing. Additionally, DOCX has largely superseded DOC as a file type.

I converted the target text to PDF with the `reportlab` Python library[19], and converted to DOCX with the `python-docx` library[20].

[15] Harvey, *Vault: Analyze Large Document Sets at Scale*, HARVEY SUPPORT (Feb 4, 2026) https://help.harvey.ai/articles/vault [https://perma.cc/75JL-7B7E] (last visited Feb. 17, 2026).

[16] Thomson Reuters, *Supported file types*, THOMSON REUTERS https://www.thomsonreuters.com/en-us/help/cocounsel/tax-audit-accounting/conversations/supported-file-types [https://perma.cc/WN77-PBBY] (last visited Feb. 17, 2026).

[17] Lexis+, *Upload Documents*, LEXISNEXIS https://help.lexisnexis.com/Flare/lexisplusai/US/en_US/Content/FAQ/upload.htm?Highlight=pdf [https://perma.cc/4MSP-CYEG] (last visited Feb. 17, 2026)

[18] LexisNexis, *File Formats and Limits*, LEXISNEXIS https://help.lexisnexis.com/Flare/casemaponline/US/en_US/Content/file_format_limit.htm [https://perma.cc/7J4S-LW6F] (last visited Feb. 17, 2026)

[19] REPORTLAB DOCS, *Andy Robinson* et al., https://docs.reportlab.com/ (last accessed March 6, 2026).

[20] PYTHON-DOCX 1.2.0 DOCUMENTATION, *Steve Canny*, https://python-docx.readthedocs.io/en/latest/ https://perma.cc/RG7D-UT4K (last accessed March 6, 2026).

3.1.2.4 File Set Sizes

This research used file sets generated with 10, 25, 50, 100, 250, and 500 files. Although there were initially ambitions to test the full 100gb of vault space that Harvey offers, Lexis caps the number of files in a vault at 500. In order to make the comparison between vendors comparable, the number of documents I used in this study was capped at 500 for all legal AI programs. I chose six file set sizes spanning this range. I chose these specific sizes because within these bounds, they were round numbers which roughly followed a logarithmic progression that convered the tested range.

I created a new set of files for every corpus, at every file set size, and every trial. This individual file set would be uploaded to all three legal AI platforms. E.g. the files generated for trial 3 of the size-100 Markov corpus would be uploaded to all three tested legal AI platforms, but the files for trial 4 would be different. After creating a vault/database, they were queried once and never altered.

3.2 Clues

In order to ensure regularity and fairness of the questions given to the legal AIs, I developed a standard set of clues and questions to spread across all corpora and ask all platforms. These were then compared to a corresponding set of answers. A clue was a set of clues were statements injected into files in the file set. Questions are the queries I asked the legal AI based on the file set containing the clues. Answers are the legal AI's response.

This study evaluates the ability of legal AI tools to find these specified clues rather than inquiring about the actual underlying corpora documents. Without doing this, results would likely have been difficult, if not impossible, to compare from corpora to corpora due to differing baseline performance from questions based on the corpora. For instance, the difficulty of answer a question about an email versus a contract or is difficult to quantify. While such a study may yield insightful data, it would come at the cost of directly comparing findings from one corpora to another.

All clue templates and fillers are available at the linked repository.

3.2.1 Clue Creation

I created clues by first creating templates for retrieval or logical problems. Then for each file set, I randomly chose retrieval and logical problems from the templates, and populated in random names of people, reports, facts, topics, and randomly selected dates. These became the "clues" that I injected into the file set. To avoid interference between sets of clues, all files in the file set were first checked for the names that would be used. If there were no conflicts, a name would be used once and never repeated within a set of clues outside of that clue/question combination.

I chose three types of problems: simple retrieval, formal logic, and informal logic. These roughly simulate use cases which may arise from use of these tools in a legal context. Two problems of each type were included in each file set, totalling six clues per file set. One question for each set of clues was then asked to the AI tool on the basis of the uploaded file set. The legal AI's response was then compared with the answer for that problem. A restriction to six problems per file set is a limitation of this study. I chose this limit because although six sets of problems is very sparse for a file set of size 500, more than half the files could have gotten a clue with a file set size of 10. I decided not to scale the number of sets of clues as file set sizes scaled so that the data could reflect the legal AI's ability to find information which was increasingly diluted. Further research may seek to study legal AI performance if the number of problems sets did scale.

The full templates, as well as the populated names are available at the Github Repository.

3.2.1.1 Simple Retrieval

Simple Retrieval problems required the legal AI to find the literal, single-clue answer to a question. For instance, a template was:

```
"{x} met with {y} on {date}."
```

With a question an answer such as:

```
question: "When did {y} meet with {x}?"
answer: "{y} met with {x} on {date}"
```

3.2.1.2 Formal Logic

Formal Logic problems required the legal tech product to make a strictly formal logical deduction. These relied on the rules of inference for statement logic, specifically Modus Ponens, Modus Tollens, Conjunctive Syllogism, Disjunctive Syllogism, and Hypothetical Syllogism.

For instance, a Formal Logic template was:

```
"{x} and {y} do not both know of {topic}."
"{y} knows of {topic}."
```

With a question and answer such as:

```
question: "Does {x} know of {topic}?"
answer: "no"
```

3.2.1.3 Informal Logic

Informal Logic problems required the legal AI to make an informal logical deduction. These often dealt with probable or constructive knowledge. For instance, an Informal Logic template was:

```
"There is substantial evidence proving {topic}."
"{x} considers all available evidence when making assessments."
```

With a question and answer such as:

```
question: "Does {x} believe {topic}?",
answer: "Likely"
```

Note that these questions intentionally incorporate an element of probability and uncertainty and subjectivity. This intentionally mimics queries which may require this kind of reasoning in legal use cases where available information may not be formulated in complete sets of formal statements.

3.2.2 Clue Injection

After completing template population, I injected the clues at random locations in random files in the file set. These often ended up in the middle of a paragraph or sentence, but could have been anywhere. Each clue was preceded and followed by two new line characters. If there were multiple statements in a set of clues, these were not kept together, and would likely end up across different files. For instance, the formal and informal logic examples given above are both two statements. Scattering and randomizing statements poses an additional challenge for both human and machine readers of documents, but it is exactly that challenge that this research poses and studies.

Clue sets were consistent for each trial of a given file set size and a given corpus for all legal AI tools, but all different trials had different clues. For instance, the files and problems generated for trial 7 of the Contracts corpus in file set size 250 was uploaded to all three vendors, which were also asked the same questions.

3.3 Experimental Trials and Evaluation

3.3.1 Uploading Files

I followed a platform-specific procedure to upload files to each platform. Each of them have an upload/processing status indicator, and for each platform, I waited until all visual indicators of upload/processing showed completion.

3.3.1.1 Lexis Protégé

I uploaded the file sets via the web UI by clicking "Documents", then "Create Vault", and uploaded with drag-and-drop. Once the upload was complete, the questions were pasted into the chat box. The questions were accompanied by "Answer all questions but DO NOT do a document by document analysis for ANY part of the response. DO NOT make a timeline." This added statement was largely successful in stopping Lexis from making a timeline.[21] Lexis' legal AI also sometimes failed to answer all questions in as given query. When this occured, I re-asked the questions that were unanswered.q

3.3.1.2 Westlaw CoCounsel

I uploaded files via the web UI by clicking "Databases" in the bottom left-hand side bar then "Create new Database" and uploaded with drag-and-drop. Once the upload was complete, I opened a new chat session and selected the files in the database with either the "Search a database (skill)" or "Files from a database" option.

In addition to the questions, I added the following instructions to the query: "Answer all questions but DO NOT do a document by document analysis for ANY part of the response. DO NOT make a timeline.". Despite this, CoCounsel often attempted to give a synopsis of each document and its relevance to the query. Since the clues were so sparse, this led to many generated responses which were mostly stating that a certain file has no relevance to a question.

3.3.1.3 Harvey

I uploaded the files via the web UI by clicking the "+" next to "Vault" in the left-hand sidebar, and uploading with drag-and-drop. With that vault open, I then pasted the questions into the chat box and selected all files in that vault when prompted. No additional instructions were added for Harvey. If Harvey asked whether it should make a table, I always selected no.

3.3.2 Trials

I attempted ten trials for each of the three Legal AI tools at each of the six sizes of file sets, for each of the five types of corpora. In total, I created 900 total vaults/databases, asked and graded 5,400 questions, and uploaded a total of 140,250 files amounting to 29.43 GB of information.

Note that the objective of these trials was not to produce an ideal response, but a realistic one. Although each set of questions for each file set could have been retried repeatedly to get a better result, my priority was to mimic a workflow of someone simply using the tool. Better custom prompts/information may also have yielded better performance for any given legal AI product, but developing custom instructions or iteratively improving prompts represent a different use case than the one modeled in this study.

Note that for reasons relating to the author's available hours, I often uploaded larger file sets then left them overnight for the platforms to "process" the files.

3.4 Evaluating outputs

Since I injected six clues into each file set and asked six questions to each legal AI, retrieval capabilities were graded out of a full score of six. All questions were graded by the author.

Especially with informal logic questions, responses were still recorded as correct if the AI found all the clues and related them to each other with a logical framework even if the overall conclusion was contrary to the prescribed answer. For instance, a formal logic (hypothetical syllogism) template was:

```
"If {x} has the email about {topic}, they would have shared it with {y}.",
"If {y} has email about {topic}, they would have shared it with {z}."
```

The question would then ask:

```
"If {x} has the email about {topic}, would {z} have email about {topic}?",
```

Answers I recorded as a success include answer such as:

[21] It is unclear to me what triggers Lexis Protégé to make a timeline. Sometimes when question sets included questions about dates (e.g. `When did {x} learn about {report}?`) Protégé would try to make a timeline, but inconsistently. The custom instructions were required to curtain this unexpected behavior as much as possible.

```
"No, because although there is a hypothetical chain of custody from {x} to {z},
    there is no direct evidence showing that {z} has the email about {topic}"

"There is no conclusive evidence that if {x} has the email about {topic}, {z} would have it.
    There are only statements saying that if {x} has the email about {topic},
    he would have given it to {y}, who would have given it to {z}."
```

In instances like this, since the legal AI retrieved the relevant statements, I recorded this as a success even though the conclusion is opposite to the formally logical conclusion. This flexible standard and consideration was intentional to focus on the retrieval properties of the legal AI, not its ability to do the more human work of evaluating whether evidence meets a certain standard of conclusiveness or certainty.

Although formal and informal logical reasoning serve as some of the templates of clues and questions for this research, information retrieval, and not the ability to adhere to the precise conventions of statement logic, is the focus of the research. If a legal researcher found such an answer as the examples above, the answers would still be useful to them since it identifies the clues were found. The Legal AI's responses also often linked to locations in the underlying documents.

The raw data of recorded scores is available in the linked Github repository.

4 Findings

4.1 Refusals

In some instances, the trial would not result in recordable data. If less than 50% of trials at a given file set size for a given corpus for an legal AI product yielded recordable data, I report no data for that corpus at that file set size for that legal AI product. This threshold was chosen to balance usable data and maintain integrity among trials.

With even a 10-file file set, Lexis often produced a fatal number of errors when uploading files of Random and Zero file sets. This may have been due to a content filter or pre-processing function built into the upload tool. At 500-file file sets, Lexis would also return errors instead of responses to the given query for Markov file sets. Accordingly, no recorded data exists for Lexis for Random or Zeros, and results for Markov at file set size 500 was not recorded.

Although Westlaw CoCounsel allowed the upload of files for all corpora and at all file set sizes, the CoCounsel legal AI itself, instead of the document upload function, often returned a query-time error with the Random corpus at larger sizes of file sets. Accordingly, no data is recorded for the Random file set sizes 100, 250, and 500 for Westlaw.

I ended up recording the following data points for each legal AI:

Legal AI	Responses
Lexis	175
Westlaw	268
Harvey	300

4.2 Correlation between File Set Size and Legal AI Performance

There is some correlation between file set size and performance for some legal AI tools. After preliminary data exploration, the Exponential Decay function seemed to best describe the data.

The Exponential Decay function is defined as:

$$y = ae^{-bx} + c$$

Because I have multiple data points across different trials for any given file set size, I fit the Exponential Decay function to weighted mean responses at any given x value in order to show the underlying trend. This function fit is simply a descriptive model, not an inferential claim, and therefore does not use hypothesis testing.

In order to fit models to this data, I found a weighted averaged of every file set size across all corpora. This was the y value, and the file set size was the x. I then fit these points with the `curve_fit` function in the `scipy.optimize` library.[22]

Vendor	R^2	MAE	a	b	c	se(a)	se(b)	se(c)
Harvey	0.051549	0.66077	0.63726	0.017073	4.7518	0.57980	0.036961	0.30770
Lexis	0.50643	0.79910	3.2071	0.013462	0.97018	0.85813	0.0093893	0.53882
Westlaw	0.44218	0.58271	2.8971	0.0024473	2.4232	2.4689	0.0038972	2.6234

Note that these values are calculated on the raw data, not percentized scores, which is what is represented on the following visualizations.

4.2.1 Lexis and Westlaw

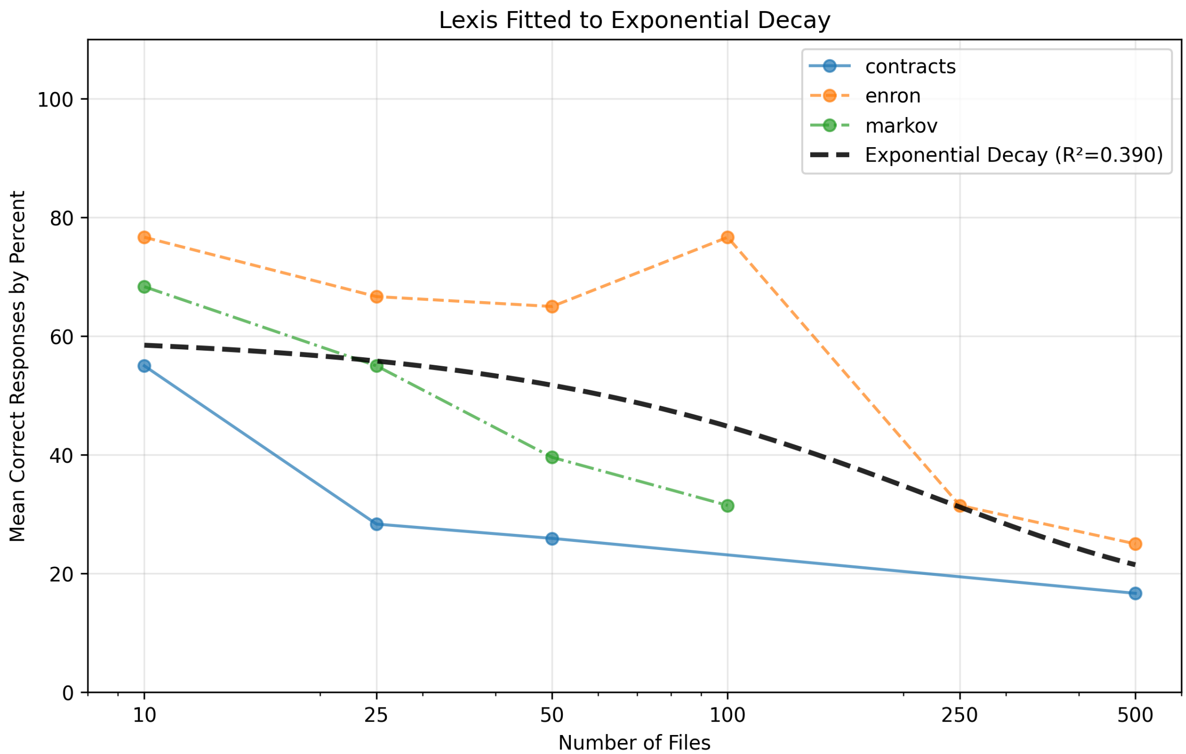

Figure 1: Lexis means by corp

As a goodness-of-fit measure, the R^2 of Lexis (0.506) and Westlaw (0.442) shows that the exponential decay function meaningfully describes a trend in the data, explaining ~51% and ~44% of the variance in the data respectively. While the exponential decay function is descriptive of the collected data, there is also variability in the data. This R^2 is influenced by a number of factors. Because file set sizes scale approximately logarithmically, gaps between file set sizes (x-values) are likewise spaced logarithmically. These increasing gaps between recorded x-values can make function fitting a challenge. Despite this challenge, exponential decay is still clearly observable and describable within the x-value range.

Additionally, this function is fit to data averaged from all corpora for each given file set size. In reality, it is likely that there are different parameters for different corpora, and that modeling a function on each

[22] Scipy, *scipy.optimize.curve_fit*, SCIPY (2008) https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.curve_fit.html [https://perma.cc/RE8X-EKGA] (last visited Feb. 17, 2026)

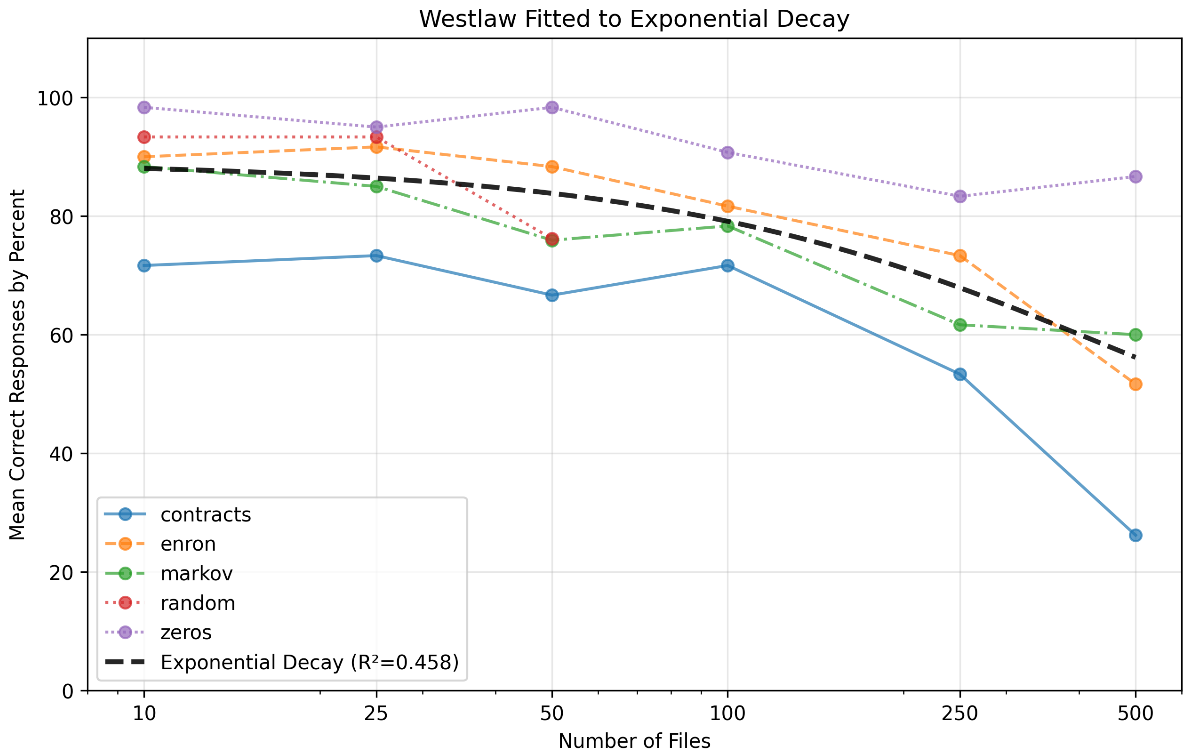

Figure 2: Westlaw means by corp

corpus individually either with custom parameters ($y = a_{corpus}e^{-b_{corpus}x} + c_{corpus}$) or adding a delta offset for each corpus ($y = ae^{-bx} + c + \delta_{corpus}$) would yield a higher R^2 value. While this is an interesting area for further research, this would likely require much more data-gathering and is outside the scope of this study to record data at a volume that is statistically significant.

Although the primary focus of this model fitting is descriptive and not predictive, Mean Absolute Error (MAE) describes the average amount that predictions made on the basis of this model would be off by. Interpretively, the given amounts show that in general, MAE shows that Westlaw's responses generally showed less variation in its mean responses, whereas Lexis's responses had more mean variation.

Note that due to its errors and refusals, the curve describing Lexis' performance was fit on less data from Lexis than the other legal AI tools.

4.2.2 Harvey

Unlike Lexis and Westlaw which fit to a exponential decay function with statistically representative R-Values, data exploration found that no function properly fit the data from Harvey. The line shown in the graph shows the best-fitting exponential decay curve, but it fits the data very poorly ($R^2 = 0.051$), and looks very linear. This line is visually included to show what may be an averaged overall trend, not to indicate fit. There is no real relationship between the fit set size and performance that fit a regression model.

In lieu of an appropriately representative model, this study instead can only describe Harvey's performance across all file set sizes as centering around 83.3% ± 10% when all corpora are averaged (range: 73.3% - 91.7%).

4.3 Correlation between Corpora and Legal AI Performance

There is correlation between the corpora and the Legal AI performance.

I performed Analysis of Variance (ANOVA) on all data points for each corpora to determine whether

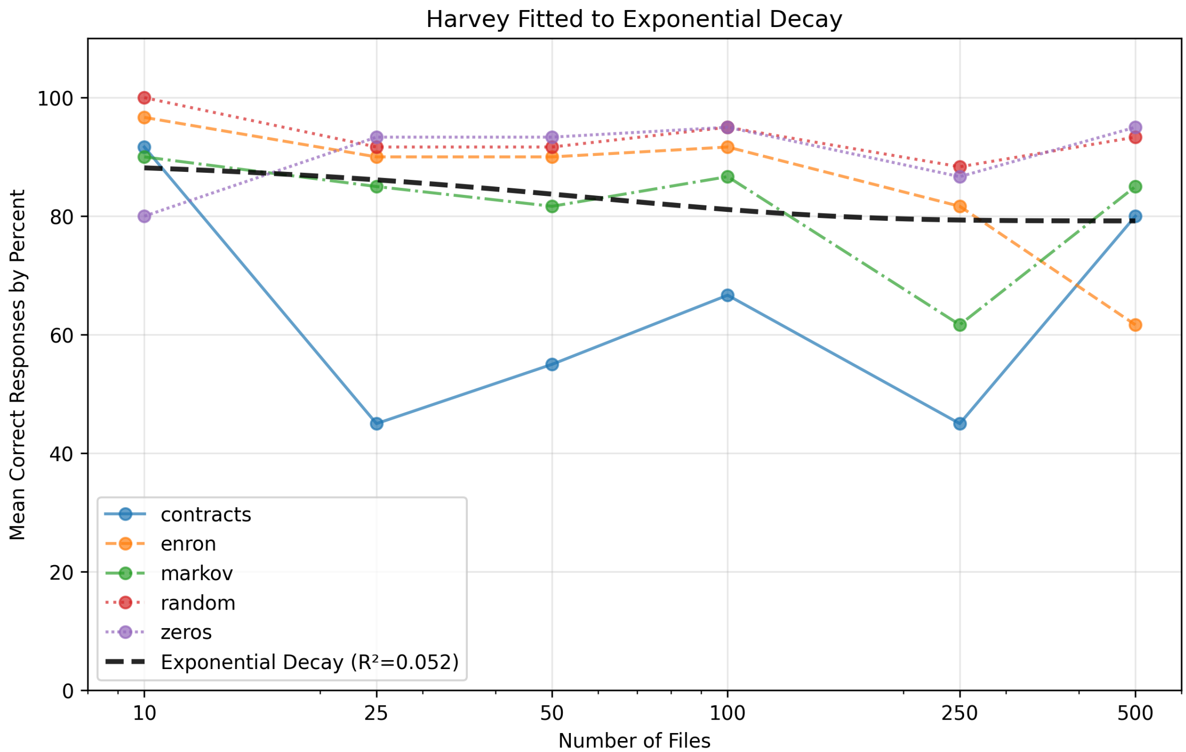

Figure 3: Harvey means by corp

performance of different corpora are statistically distinct, i.e. whether the variance between groups was greater than variation within groups. The Null Hypothesis was that performance for every corpus was the same:

$H0 : contracts = enron = markov = random = zeros$

with the alternative hypothesis stated as:

$Halt$: it is not the case that $contracts = enron = markov = random = zeros$

ANOVA yielded a P-value of 3.2642672702541746e-46, far smaller than any conventional significance level. This means there is extremely strong evidence that at one or several groups differ from other groups.

I followed this with a variation of Tukey's Honestly Significantly Different test called Tukey-Kramer. This variation is used when different data labels had different numbers of data points. This test determines which pairs of groups were statistically distinct from each other.

Tukey-Kramer yielded the following analysis for each pair of corpora compared against a 5% significance value:

Group 1	Group 2	dij	HSD	Reject?
contracts	enron	1.1907	0.40118	True
contracts	markov	0.90326	0.41037	True
contracts	random	2.2610	0.49120	True
contracts	zeros	2.2227	0.44637	True
enron	markov	0.28742	0.39923	False
enron	random	1.0703	0.48193	True
enron	zeros	1.0321	0.43615	True
markov	random	1.3577	0.48960	True
markov	zeros	1.3195	0.44462	True
random	zeros	0.038250	0.52015	False

For each pair of groups, the Null Hypothesis is:
$H_0 = group1 = group2$,
and the alternative hypothesis is
$H_{alt} =$ It is not the case that $group1 = group2$.

The "Reject" column describes whether on the basis of the Tukey-Kramer methodology, the Null hypothesis is rejected and the alternative hypothesis is adopted. Here, the alternative hypothesis is adopted for all pairs except Enron with Markov, and Random with Zeros. This means that according to Tukey-Kramer, the variation within each group in a pair does not exceed the variation between the groups in the pair. For all other groups, the distinction is statistically significant.

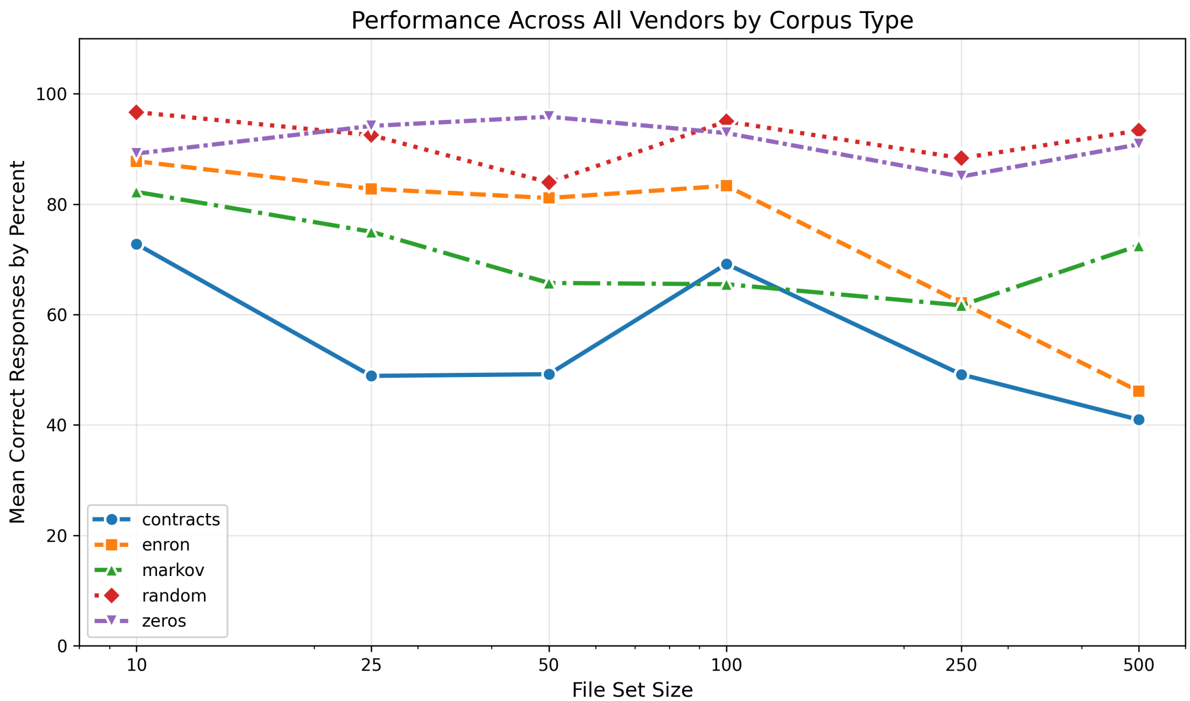

Figure 4: Corpora performance

After determining that there is statistical difference overall as well as most pairs of groups, I compare these differences. The best recorded performances came from Zeros and Random. Although note that this data may be skewed because data is missing from Lexis for these corpora. After Zeros and Random, the next corpora, in order of performance, is Enron, then Markov, followed by Contracts.

The least performant corpus was Contracts. This result is surprising because analysis of large volumes of contracts is sometimes an explicitly advertised use of these legal AI tools. The poor performance may be due to the density of information in the original underlying corpus. Perhaps because of the high volume and density of legally significant semantic content in the underlying data, all legal AI tools seemed to suffer significantly in terms of performance. At nearly every file size, nearly all vendors performed the worst with the Contracts corpora.

This may indicate an inherent challenge for retrieval based legal AI tools. Retrieval architectures pair a set of documents with a large language model which can use it as a basis of knowledge that is outside of its context window and outside of the information represented in its training corpus. Retrieval architectures such as Retrieval Augmented Generation is widely used in many legal AI architectures to increase accuracy[23], sometimes even leading to claims that such such systems enable a legal AI to generate a response that is

[23] see e.g. James Ju, *Retrieval-augmented generation in legal tech*, Thomson Reuters Blog (December 4, 2024), https://legal.thomsonreuters.com/blog/retrieval-augmented-generation-in-legal-tech/ [https://perma.cc/CM78-KFSB].

"hallucation free"[24]. Such claims have been[25] and should be critically examined.

The order of legal AI performance by corpora, interestingly, seems to follow roughly the expected performance that a human would have if presented with the same tasks. Clues among corpora that are easy to dismiss as meaningless (Zeros and Random) had the best results, and information containing more cognizable and semantically meaningful information yielded worse performance.

5 Discussion and Other Findings

The following observations are derived from extensive interaction with the studied legal AI platforms and provide context for statistically-determined quantitative findings.

5.1 Platform Speed

The speed of these legal AI tools varies widely at each stage. For reasons mostly unknown, Lexis was, by far, the slowest to upload files. The only information I received that may contribute to explaining why came from Lexis themselves. During my experimentation, a Lexis employee contacted me and informed me that my use load was negatively impacting "the product's overall performance"[26]. They explained that this was leading to slower upload and processing times. Although they did not ask me to stop or throttle my access, they did ask about the nature of my use to better allocate resources for it. First, this may be indicative of overall scale issues with the this legal AI in its current deployment. Second, this may explain the performance of Lexis' legal AI tool in this study. Without further information, I cannot conclusively address either issue.

Westlaw was very fast to upload files to, but running a query through the Legal AI can be extremely slow. With large file set sizes, responses often took upwards of 10 minutes.

Harvey's upload and query process was unparalleled in terms of speed and ease of use. Files uploaded quickly, had minimal errors, and the query process did not seem to take much longer on 500 files than it did on 50.

5.2 Performance Between Problem Types

One limitation of this study is that problems of different types were not individually recorded. I made this decision in order to manage the scope of the research, prioritizing more data that could hopefully be more representative of a more focused set of variables rather than less data at a higher specificity. This area is ripe for further exploration and analysis, and could have a significant impact on the ability to develop functional guidelines and best practices for use of legal AI tools.

However, from informal observation, it seemed that Simple Retrieval tasks were the easiest for all legal AI tools, and Formal Logic was the hardest.

5.3 Struggles With Non-linearity

The difficulty with Formal Logic seemed to especially stem from tasks that required non-linear retrieval. For instance, if was clue template was:

```
"{x} and {y} do not both know of {topic}.",
"{y} knows of {topic}."
```

and the corresponding question was:

```
"ques": "Does {x} know of {topic}?"
```

[24]LexisNexis, *LexisNexis Launches Lexis+ AI, a Generative AI Solution with Hallucination-Free Linked Legal Citations*, LexisNexis News (October 25, 2023), https://www.lexisnexis.com/community/pressroom/b/news/posts/lexisnexis-launches-lexis-ai-a-generative-ai-solution-with-hallucination-free-linked-legal-citations [https://perma.cc/D24V-JADS]

[25]see e.g. Varun Magesh et al., *Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools*, J. Empirical Legal Stud. 1–27 (2025), https://dho.stanford.edu/wp-content/uploads/Legal_RAG_Hallucinations.pdf

[26]Email from Adriana Ramirez, Research Attorney Practice Area Consultant, to Justin Tung, Reference Librarian and Lecturer at the Univ. Tex., (Feb 4, 2026 7:46 AM CST) (on file with author).

Reaching a conclusive answer requires the legal AI to not only find and analyze information about {x}, the name explicitly given in the answer, but also {y}, who is not named in the answer. Many times, the legal AI would simply respond that either {x} or {y} knows of the topic, showing that it likely did not make that logical step in appropriately modifying its scope to retrieve the information it used to formulate its answer.

Like differing problem types, further research into this dynamic could instrumentally impact the development of guidelines and best practices for legal AI use.

5.4 Processing Paralleliztion

Through interaction with the platform, I speculate that the consistency of Harvey's performance across different file set sizes is due to parallelization, i.e. the use of parallel processes that simultaneously spawn multiple instances of an LLM to ingest and process files from the set. This is specualtion is based on the fact that Harvey's response times seemed roughly constant across different file set sizes, whereas other legal AI response times scaled in porportion to the file set size. Although parallelization an extremely powerful technique, one drawback to this architecture is that it can lead to heavy processing demands and high computing costs. This architecture trades processing and costs for speed.

Harvey's different technique may be a reflection of a different product strategy than Lexis and Westlaw. Without the ability to directly control and query an enormous database of legal materials[27], Harvey's product may seek competitiveness on the basis prioritizing access to computing resources. Because they also do not have the cost burden of maintain such a database, they may still stay cost-competitive even though they offer a product with different capabilities.

6 Conclusion

The legal tech market often seems to have an infinite amount of AI products purporting to sell any imaginable kind of functionality. However, the majority of available information about these tools comes from marketing material and first-party sources. Without independent scrutiny and analysis, it is easy to mistake anecdotal success for statistically impactful utility.

To test the impact of the number of files and the type of content in those files on the performance of different legal AI tools, I created file sets of five corpus types at six file set sizes to test the custom database retrieval capabilities of legal AI tools from Harvey, Westlaw, and Lexis. I injected these file sets with clues and questions pulled from a standardized template which was varied enough to avoid cross-contamination. I then repeated ten total trials per condition, and measured the ability of the legal AI tool to retrieve the relevant clues required to be responsive to the question. In total, I uploaded over 140,000 files across 900 vaults.

First, to test the influence of file set size on a legal AI's performance, I found that Westlaw's performance could be meaningfully described with an exponential decay model, beginning with an average of 88.3% at a file set size of 10, and ending at 54.2% at a file set size of 500. Lexis' performance could also be meaningfully described with an exponential decay model, beginning with an average of 66.7% at at file set size of 10 and ending at 18.7% at a file set size of 500. Additionally, no recorded data was available for two entire corpora and some file set sizes for another corpora. Harvey's performance did not fit an exponential decay model, nor any other mathematical model attempted, and instead began by accurately retrieving 91.7% of answers at a file set size of 10, and 83.0% of answers at a file set size of 500. Through the range, Harvey's performance centered around 83.3%.

Second, to test the influence of different corpora on the performance of all legal AI, I used the ANOVA method to reject the Null Hypothesis that there is no statistically significant difference between groups based on their labels, then used the Tukey-Kramer test to determine in which pairs of corpora those statistically significant differences appeared. I found that with the exception of two pairs (Enron + Markov; Random + Zeros), all other pairs of corpora showed distinguishable differences, meaning that the variation between the two groups was greater than the variation within each individual group.

[27] Harvey does provide a way to integrate its platform with Lexis, but this is not part of its base product, which is the product this study evaluated.

These findings support the conclusion that the performance of legal AI tools from Westlaw and Lexis decreases as the volume of user-uploaded files increases. These findings also show that the performance of all tested legal AI tools is influenced by the type of data in the uploaded files, with some types of text yielding more accurate results than others.

The practical implications of this information are manifold. Since performance increases with smaller file set sizes, users of legal AI products may focus on reducing the amount of uploaded information when possible to maximize performance when using Westlaw or Lexis. There may also be a file set size threshold with Westlaw and Lexis beyond which their lack of accuracy means they should not be used. Similarly, users should be wary of the way that different types of uploaded data can lead to better or worse performance. Ironically, data with a high density of legally probative statements tend to perform worse than data that does not.

In light of these findings, a best practice may be to examine the type of data to be uploaded before using a legal AI tool, or at least to scrutinize the legal AI's output according to an awareness of that data. Additionally, some legal AI tools also show a decay in performance as the amount of uploaded material increases. Together, the findings of this study supports a view of legal AI tools which have high perform under certain conditions and low performance under others. Users of these tools ought to determine their threshold of acceptable performance for a given task to decide whether the legal AI tool is the correct tool to use.

Aggregating these use cases can also inform purchasing decisions, as the cost of the legal AI tool can be weighed against the utility of that tool, as well as compared to competing tools on the market.

This study leaves to further research the differences in kinds of questions such as rote retrieval and linearity. It also leaves to further research the effect of scaling clue density with file set sizes. Much of this research requires a larger sample size per condition in order to draw statistically significant results, which was outside this study's scope.

In the context of the ever-changing legal tech market, some AI tools and their providers claim that their product will bring powerful transformations to the practice of law. This study scrutinizes a subset of those claims, specifically relating to the ability to upload a custom database and query it with a legal AI tool.

7 Appendix 1: Mean performance by Corpus

This is a breakdown of mean performance of each individual legal AI product, with a graph for each corpus type.

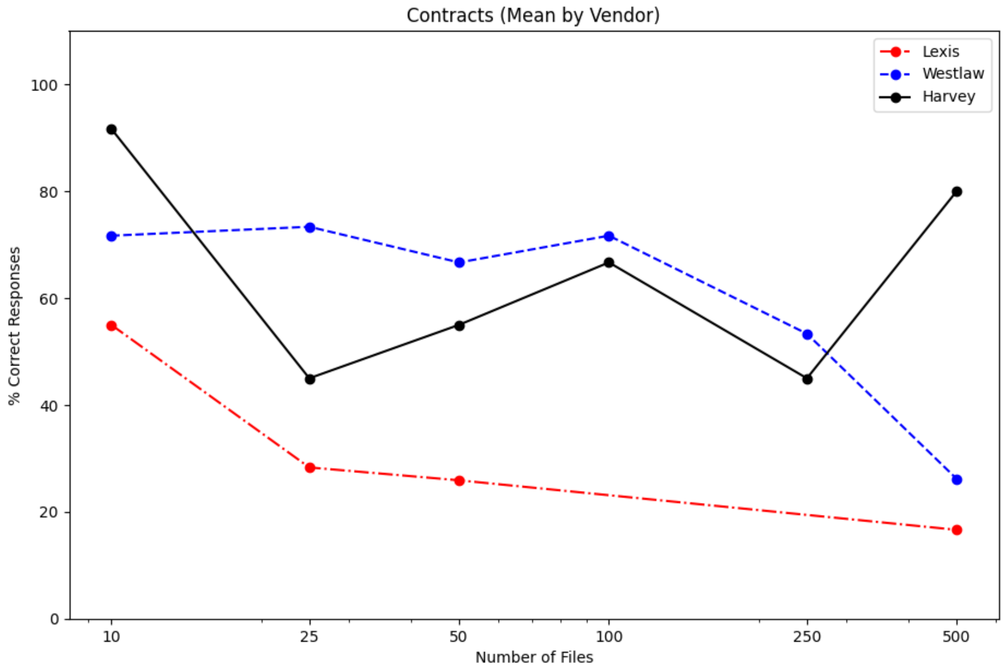

Figure 5: Contracts performance

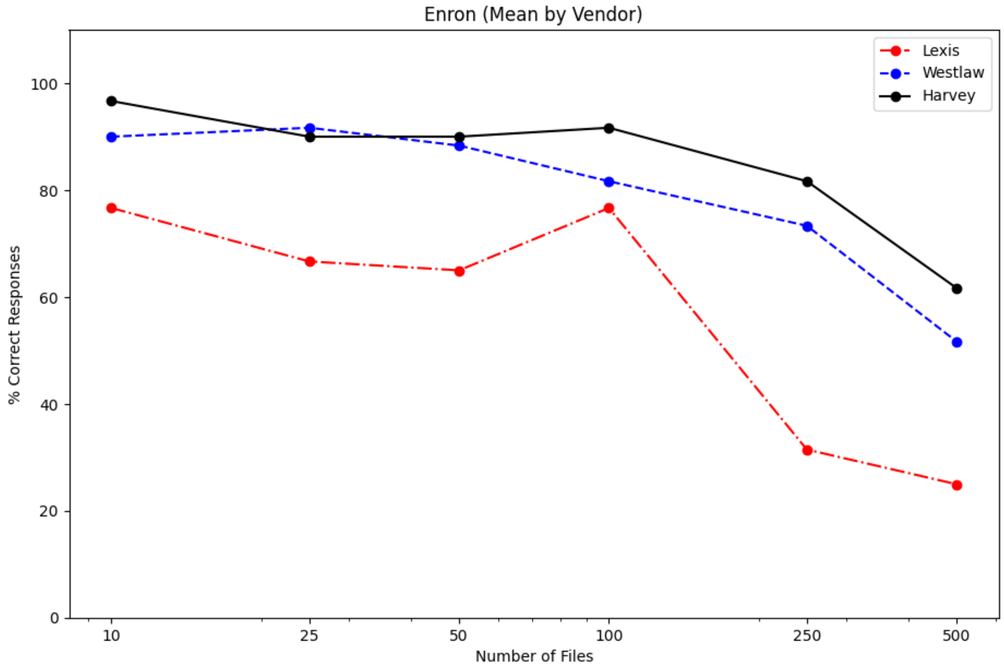

Figure 6: Enron performance

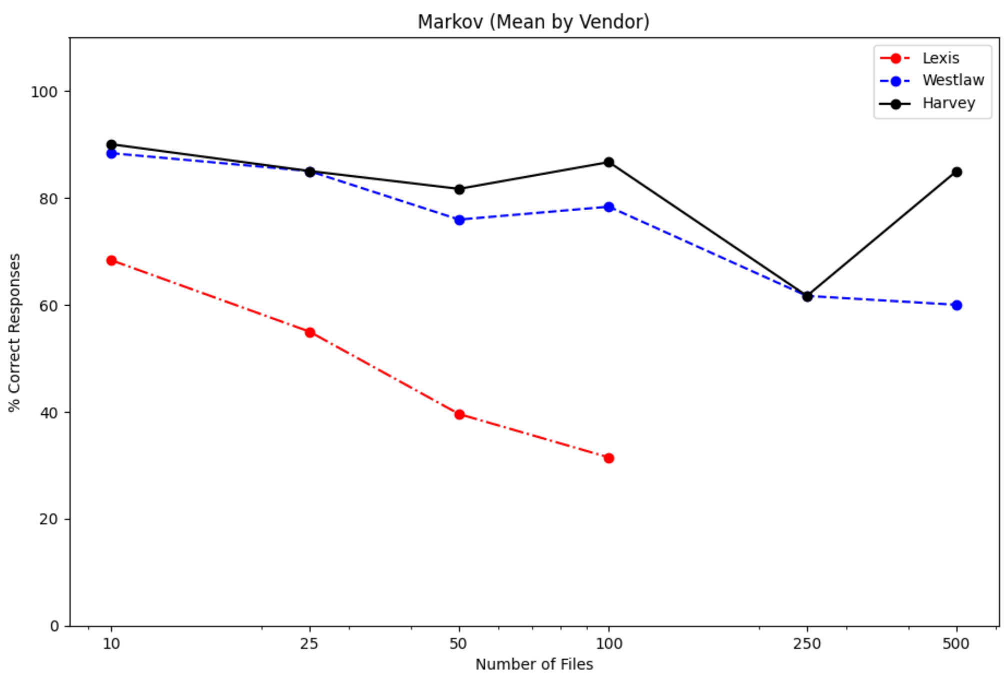

Figure 7: Markov performance

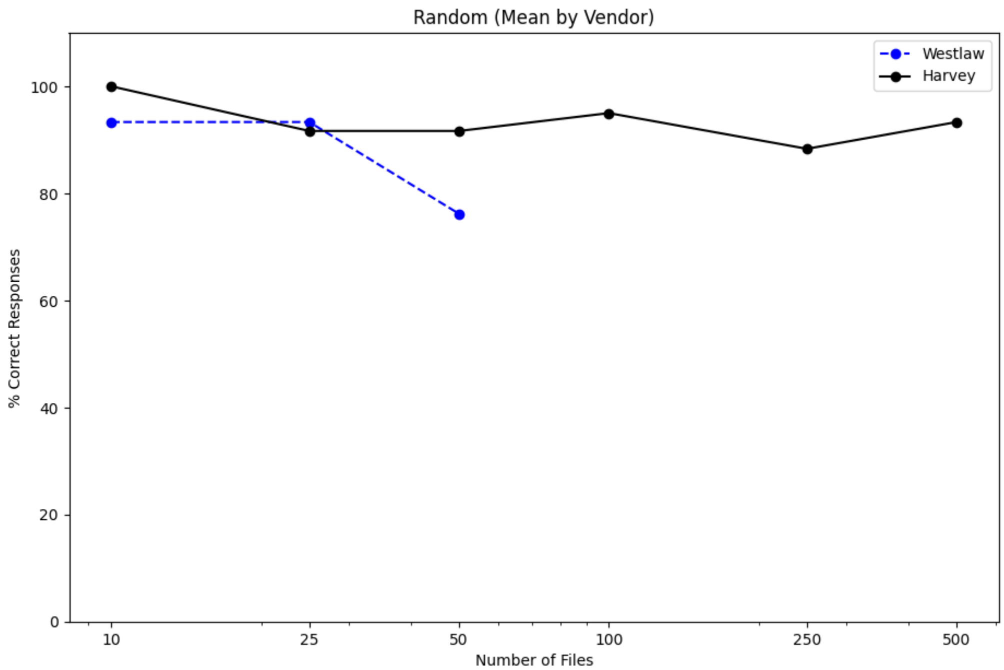

Figure 8: Random performance

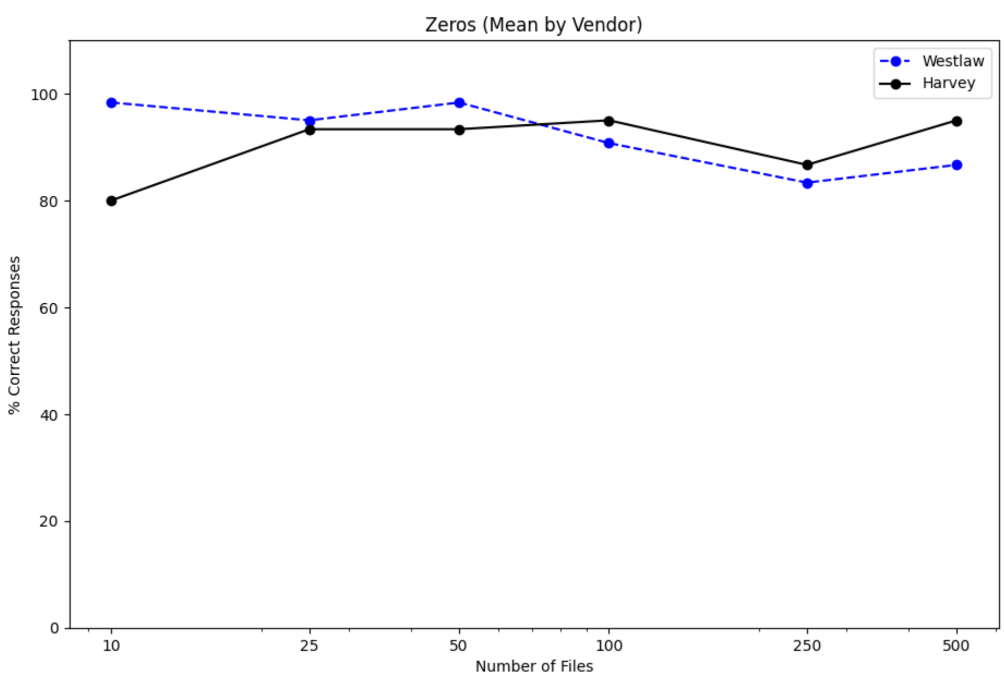

Figure 9: Zeros performance

8 Appendix 2: Findings by Quartile

This is a breakdown of the mean, 25th and 75th percentile of the distribution of responses. Every vendor with every corpus is given its own graph.

8.1 Harvey

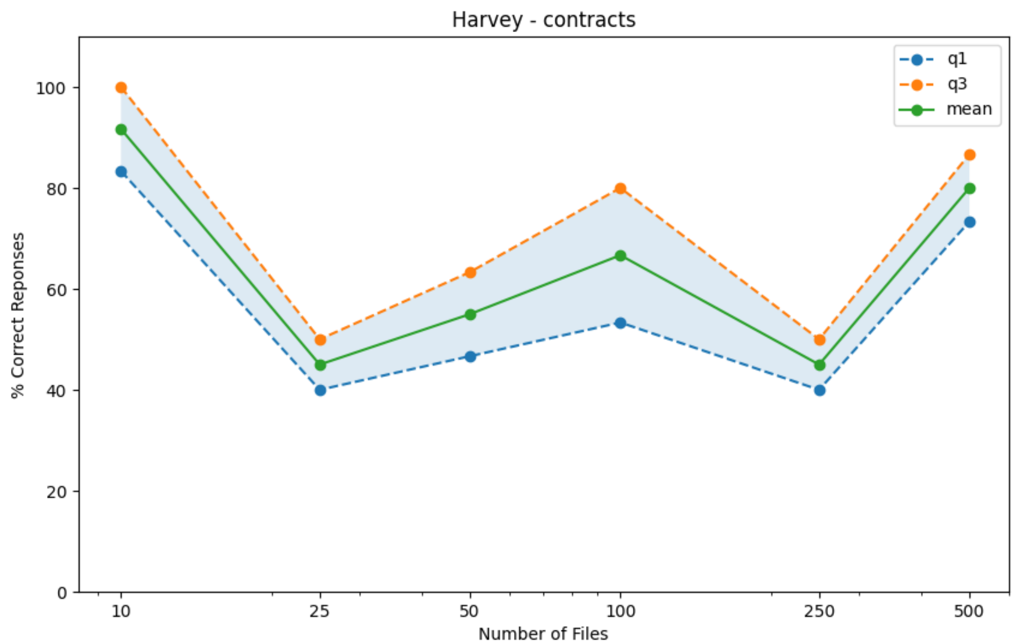

Figure 10: Harvey Contracts

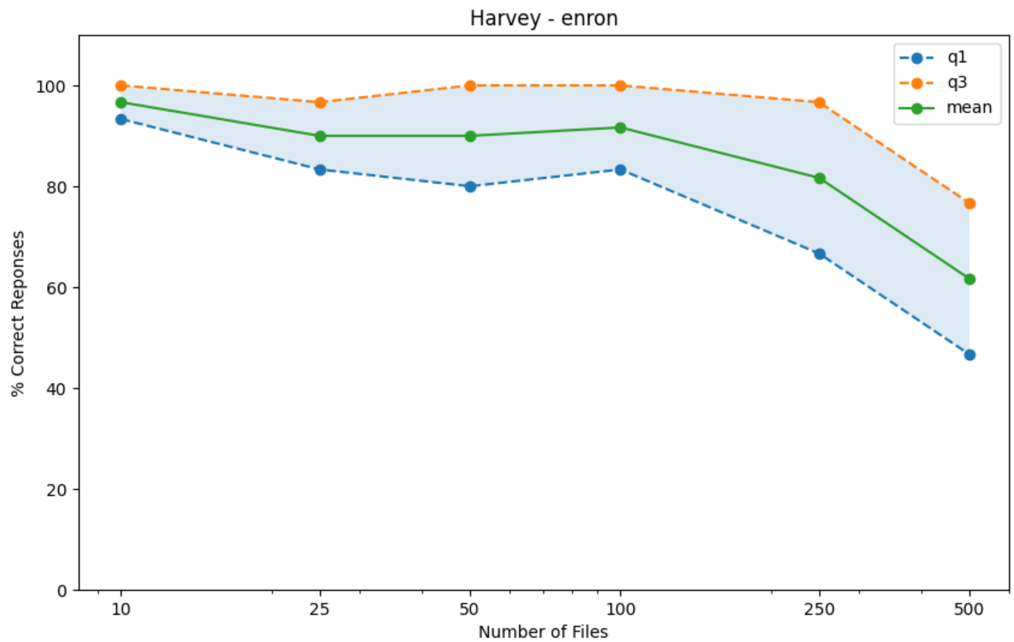

Figure 11: Harvey Enron

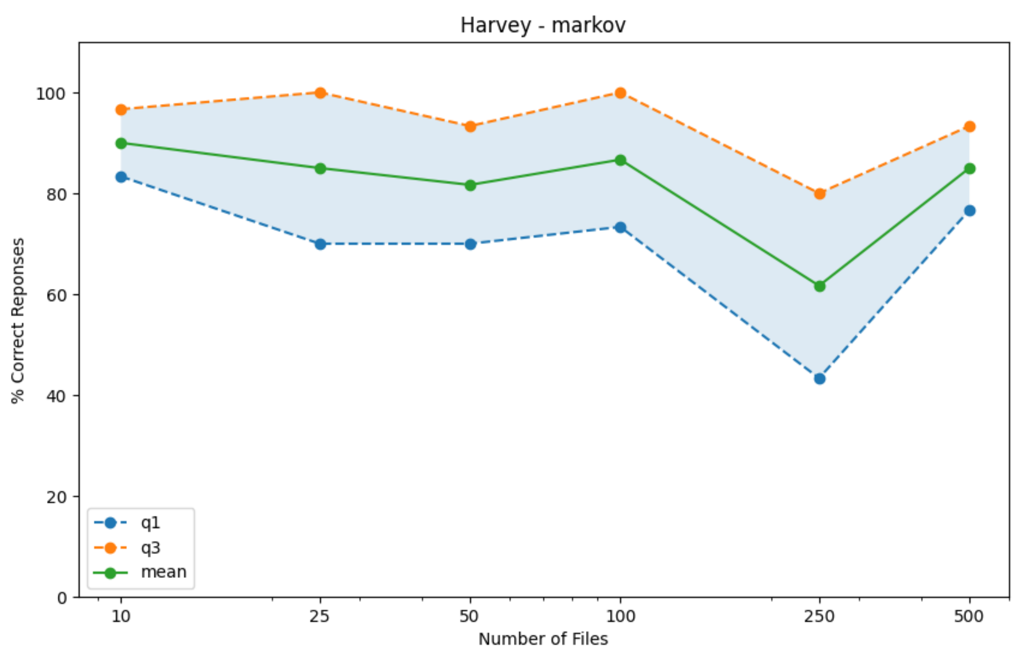

Figure 12: Harvey Markov

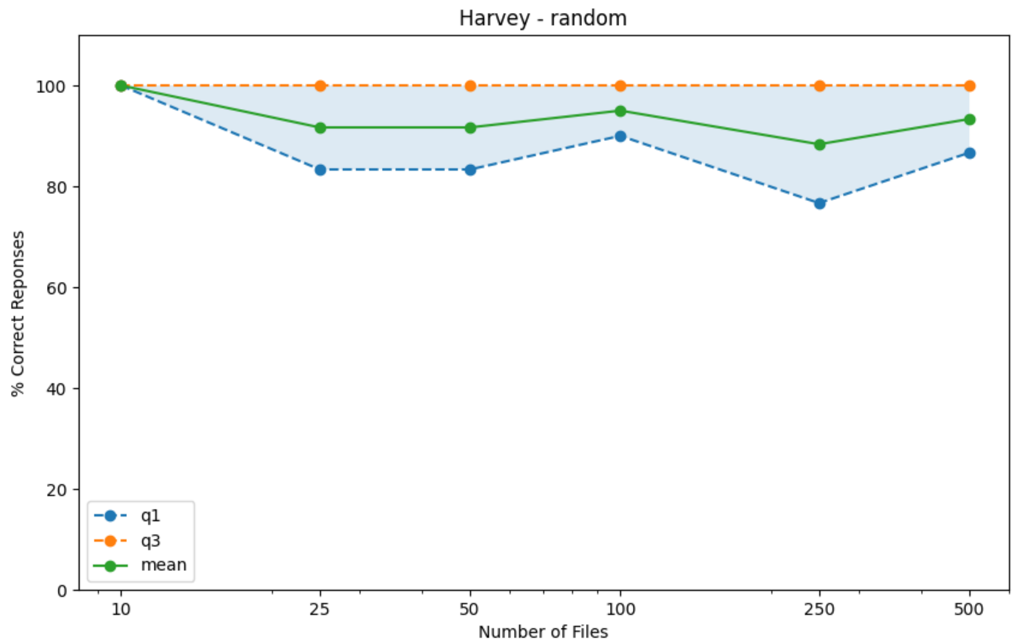

Figure 13: Harvey Random

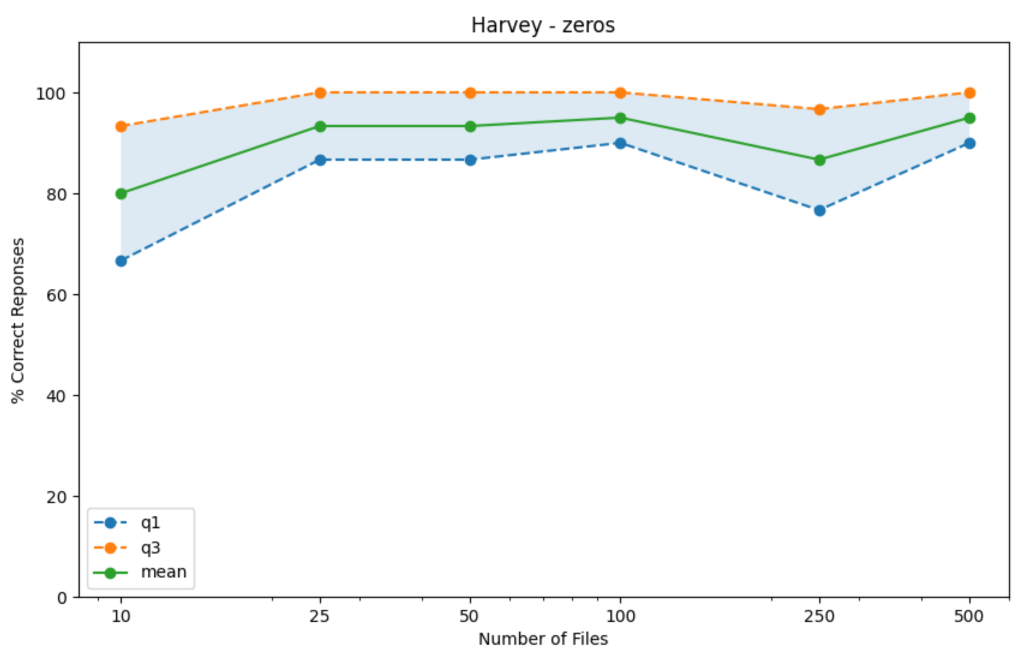

Figure 14: Harvey Zeros

8.2 Westlaw

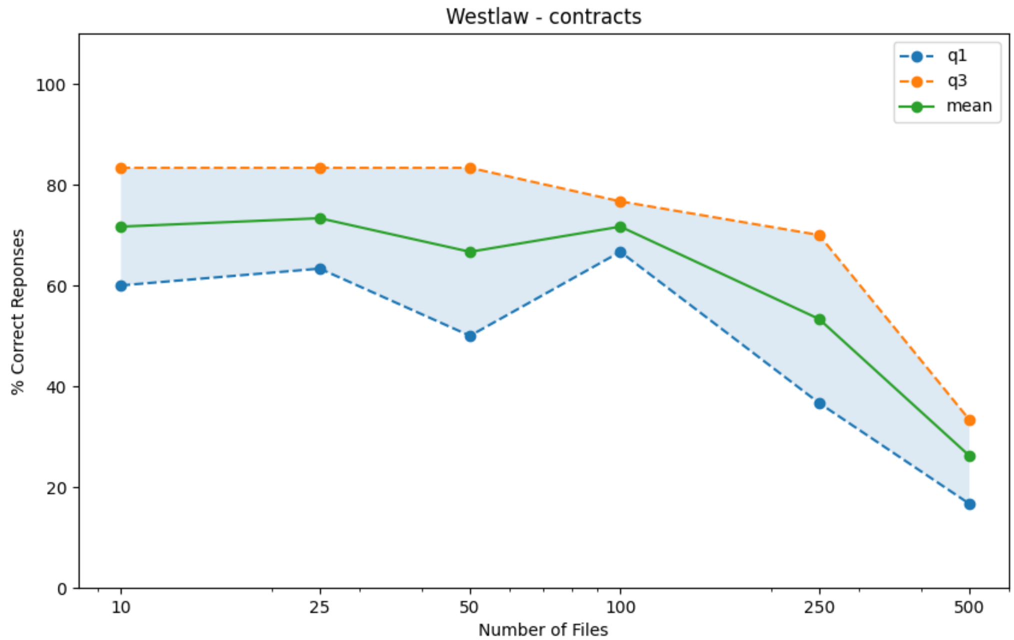

Figure 15: Westlaw Contracts

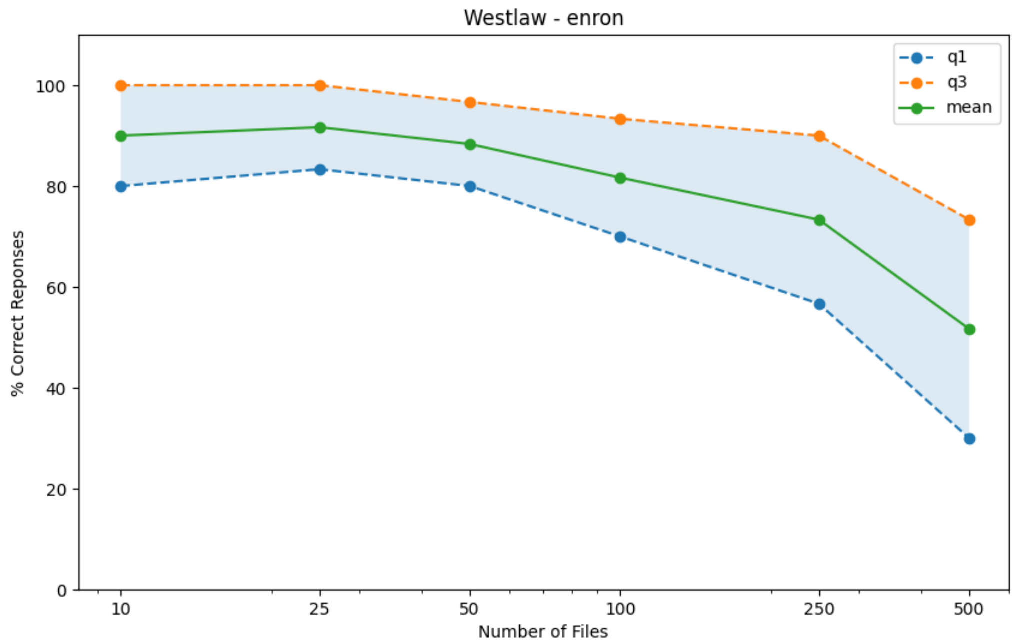

Figure 16: Westlaw Enron

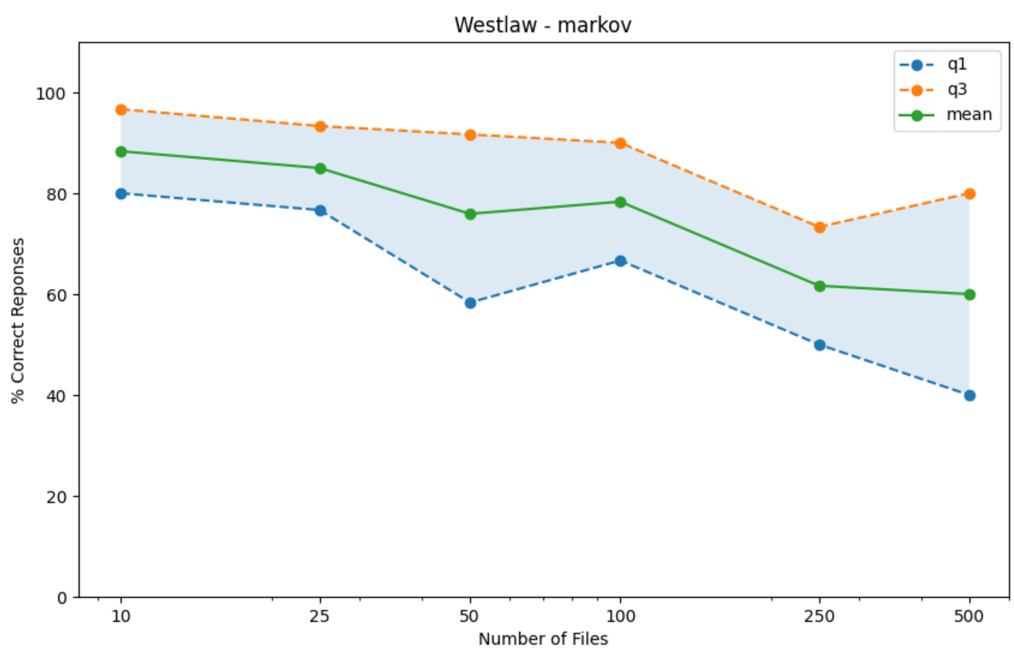

Figure 17: Westlaw Markov

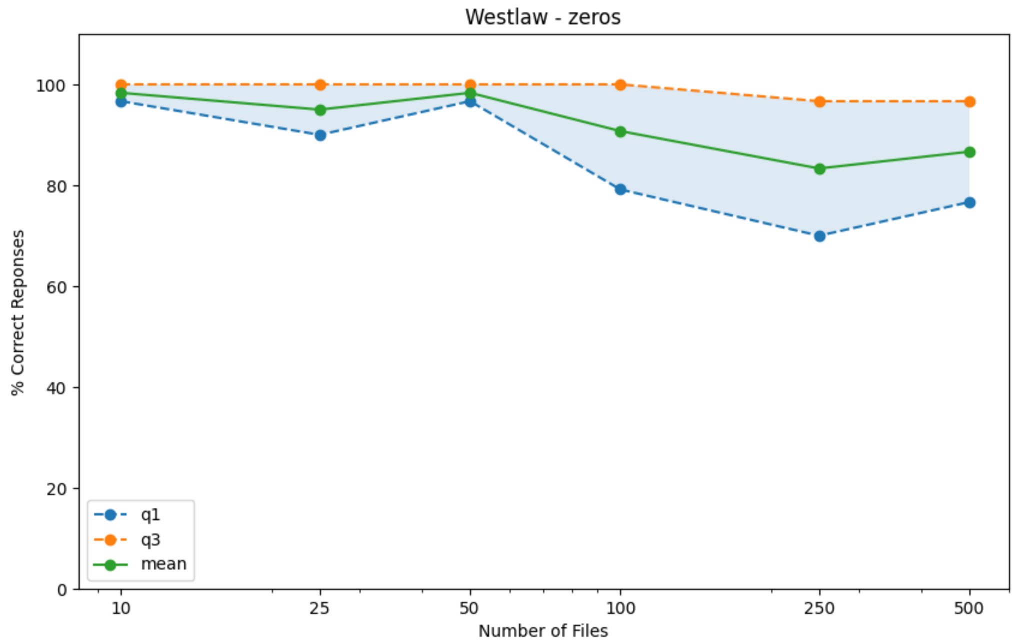

Figure 18: Westlaw Zeros

8.3 Lexis

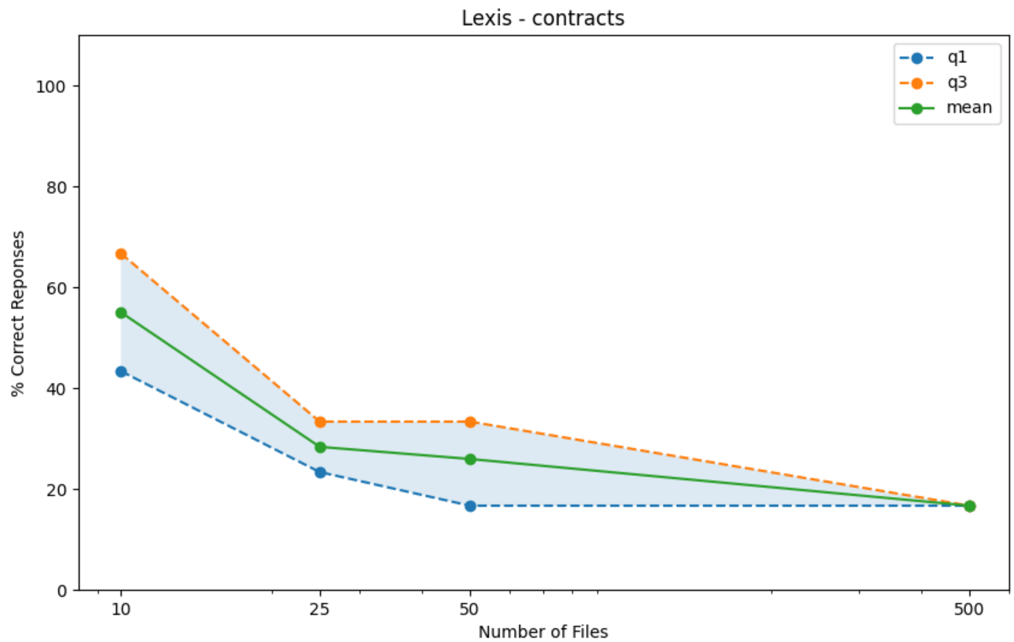

Figure 19: Lexis Contracts

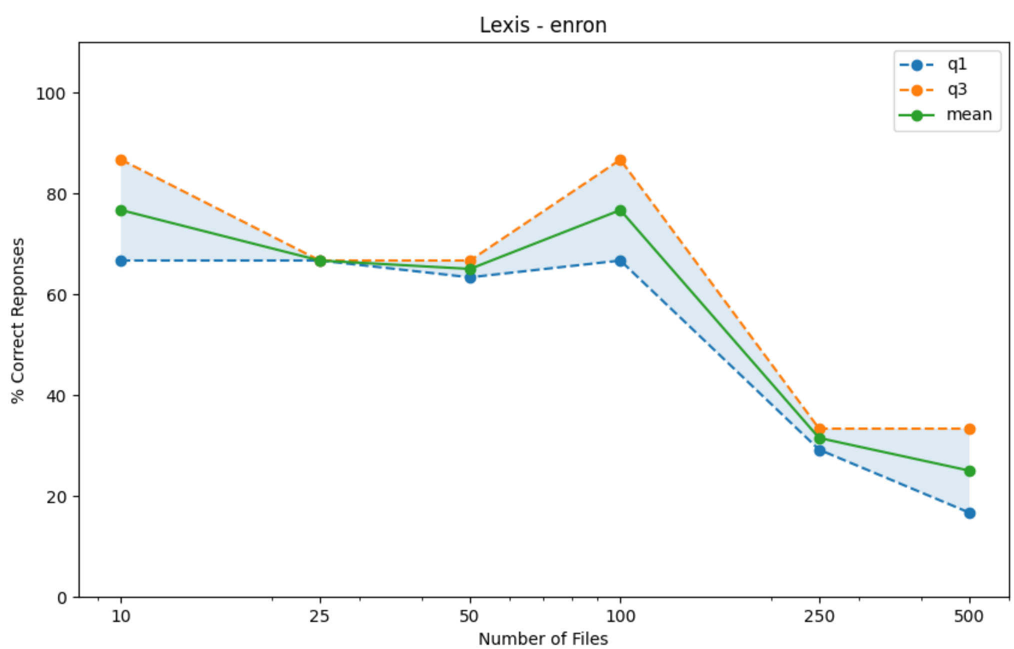

Figure 20: Lexis Enron

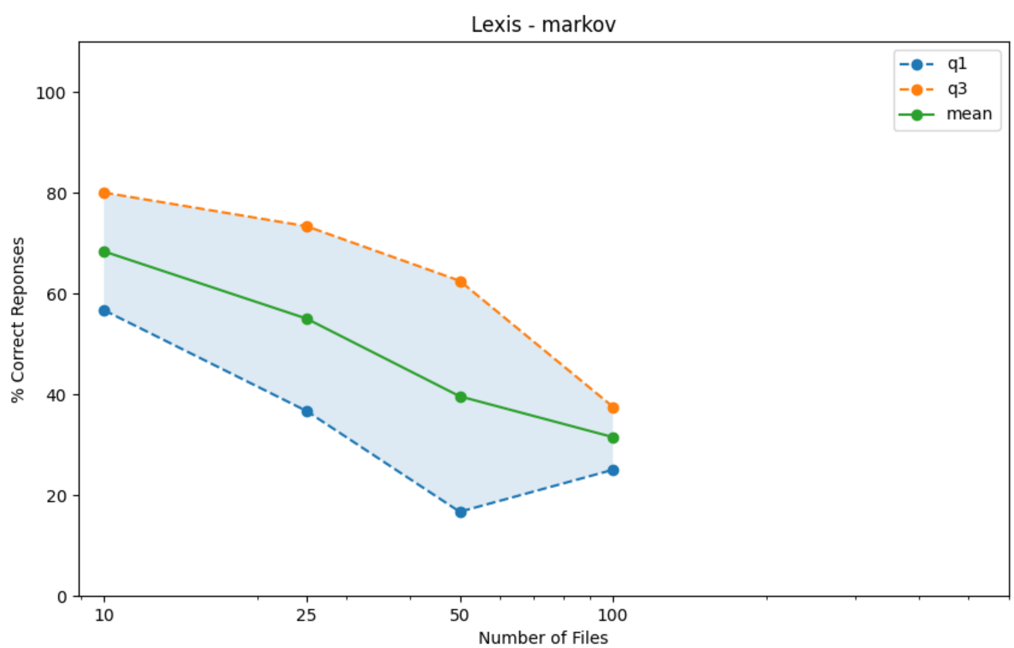

Figure 21: Lexis Markov